COMPUTER NETWORKS AND CLOUD COMPUTING

1. Which component of data communication defines rules for communication between devices?
   A) Message
   B) Protocol
   C) Medium
   D) Receiver

Answer: B

2. Which of the following is the physical path through which data travels?
   A) Protocol
   B) Medium
   C) Sender
   D) Receiver

Answer: B

3. Which topology connects all devices to a central device?
   A) Bus
   B) Ring
   C) Star
   D) Mesh

Answer: C

4. Which topology provides the highest reliability?
   A) Bus
   B) Mesh
   C) Star
   D) Ring

Answer: B

5. A network that covers a small geographic area like a classroom is called:
   A) WAN
   B) MAN
   C) LAN
   D) PAN

Answer: C

6. Which device forwards data packets between different networks?
   A) Switch
   B) Router
   C) Hub
   D) Repeater

Answer: B

7. Which device operates at the data link layer and uses MAC addresses?
   A) Switch
   B) Router
   C) Gateway
   D) Modem

Answer: A

8. Which layer of OSI model is responsible for framing?
   A) Physical layer
   B) Data link layer
   C) Network layer
   D) Transport layer

Answer: B

9. Which addressing is used at the data link layer?
   A) IP Address
   B) MAC Address
   C) Port Number
   D) URL

Answer: B

10. What is the main function of flow control?
    A) Detect errors
    B) Control data speed between sender and receiver
    C) Encrypt data
    D) Compress data

Answer: B

11. Which technique detects errors using an extra bit?
    A) CRC
    B) Parity Check

C) Checksum
D) ARQ

Answer: B

12. Which error detection technique is most reliable?
    A) Parity
    B) CRC
    C) ASCII
    D) Binary
13. What happens when a frame is lost during transmission?
    A) Data ignored
    B) Frame retransmitted
    C) System stops
    D) Channel closed
14. Which protocol requires sender to wait for acknowledgment after each frame?
    A) Stop and Wait
    B) Sliding Window
    C) CSMA
    D) Token Passing

Answer: A

15. Which protocol allows multiple frames to be sent before receiving acknowledgment?
    A) Stop and Wait
    B) Sliding Window
    C) Token Ring
    D) Ethernet

Answer: B

16. Which protocol checks if the channel is free before sending data?
    A) CSMA
    B) ARQ
    C) FTP
    D) SMTP

Answer: A

17. What does CSMA/CD stand for?
    A) Carrier Signal Media Access
    B) Carrier Sense Multiple Access with Collision Detection

C) Communication System Media Access
D) Computer Signal Management Access

Answer: B

18. Which network technology uses CSMA/CD?
    A) Ethernet
    B) WiFi
    C) Bluetooth
    D) Token Ring

Answer: A

19. In Token Ring networks, which device can transmit data?
    A) Any device
    B) Device holding the token
    C) First device
    D) Router

Answer: B

20. Why does token passing reduce collisions?
    A) Only one node transmits at a time
    B) Data compressed
    C) Data encrypted
    D) Network slowed

Answer: A

21. Which device broadcasts data to all connected devices?
    A) Switch
    B) Hub
    C) Router
    D) Bridge

Answer: B

22. Which device regenerates weak signals?
    A) Repeater
    B) Router
    C) Switch
    D) Bridge

Answer: A

23. Which device connects two similar networks?
    A) Bridge
    B) Gateway
    C) Modem
    D) Router

Answer: A

24. Which device connects different types of networks and protocols?
    A) Router
    B) Switch
    C) Gateway
    D) Hub

Answer: C

25. Which device works at the physical layer?
    A) Router
    B) Switch
    C) Repeater
    D) Bridge

Answer: C

26. Which factor determines how much data can be transmitted per second?
    A) Distance
    B) Bandwidth
    C) Noise
    D) Delay

Answer: B

27. Unwanted signals that affect communication are called:
    A) Noise
    B) Error
    C) Delay
    D) Frame

Answer: A

28. Which delay occurs due to distance between sender and receiver?
    A) Transmission delay
    B) Propagation delay

C) Processing delay
D) Queuing delay

Answer: B

29. Which technique divides a channel into multiple signals?
    A) Multiplexing
    B) Switching
    C) Routing
    D) Encoding

Answer: A

30. What is the data unit at the Data Link Layer?
    A) Bit
    B) Frame
    C) Packet
    D) Segment

Answer: B

31. What is the main advantage of Bus topology?
    A) High reliability
    B) Low cost
    C) High security
    D) High speed

Answer: B

32. How are devices connected in Ring topology?
    A) Through a central hub
    B) In a circular path
    C) Using multiple links
    D) Using wireless signals

Answer: B

33. What is the major disadvantage of Mesh topology?
    A) Low speed
    B) Limited devices
    C) High cost due to many connections
    D) Poor security

Answer: C

34. What is an advantage of Star topology?
    A) Easy troubleshooting
    B) Low bandwidth
    C) High collision rate
    D) Complex design

Answer: A

35. Which of the following is an example of a WAN?
    A) Office network
    B) Home WiFi
    C) Internet
    D) Bluetooth

Answer: C

36. Which example represents a Personal Area Network (PAN)?
    A) Internet
    B) LAN
    C) Bluetooth connection between phone and headset
    D) Campus network
37. What is the length of a MAC address?
    A) 16 bits
    B) 32 bits
    C) 48 bits
    D) 64 bits

Answer: C

38. What is the purpose of Frame Check Sequence (FCS)?
    A) Encryption
    B) Error detection
    C) Data compression
    D) Addressing

Answer: B

39. Why are sequence numbers used in frames?
    A) Increase speed
    B) Detect duplicate frames
    C) Encrypt data
    D) Reduce bandwidth

Answer: B

40. What is the purpose of the timeout mechanism?
    A) Reduce noise
    B) Detect lost frames
    C) Increase bandwidth
    D) Compress data

Answer: B

41. What is the main benefit of the Sliding Window protocol?
    A) Increased transmission efficiency
    B) Reduced security
    C) Reduced bandwidth
    D) Increased collisions

Answer: A

42. When does a collision occur in a network?
    A) When devices are idle
    B) When two devices transmit simultaneously
    C) When bandwidth increases
    D) When frames are encrypted

Answer: B

43. Why does a switch reduce network collisions?
    A) It slows down transmission
    B) It forwards frames only to the destination port
    C) It broadcasts data to all devices
    D) It removes cables

Answer: B

44. Why does a hub increase collisions in a network?
    A) It encrypts data
    B) It broadcasts data to all ports
    C) It reduces bandwidth
    D) It uses MAC addresses

Answer: B

45. What does a bridge do in a network?
    A) Connects networks using IP addresses
    B) Divides network into segments

C) Converts signals
D) Encrypts data

Answer: B

46. What is the purpose of token passing in networks?
    A) Increase noise
    B) Prevent collisions
    C) Reduce bandwidth
    D) Increase data size

Answer: B

47. Which technique is used for error control in data communication?
    A) FTP
    B) ARQ
    C) HTTP
    D) DNS

Answer: B

48. What is the formula for calculating the number of links in a full mesh network?
    A) n^2
    B) $n(n-1)/2$
    C) $n+1$
    D) $n-1$

Answer: B

49. Which standard defines Ethernet networks?
    A) IEEE 802.1
    B) IEEE 802.3
    C) IEEE 802.5
    D) IEEE 802.11

Answer: B

50. Which standard defines WiFi networks?
    A) IEEE 802.3
    B) IEEE 802.11
    C) IEEE 802.5
    D) IEEE 802.15

Answer: B

51. What information is stored in the frame header?
    A) Error detection bits
    B) Addressing information
    C) Data encryption key
    D) File size

Answer: B

52. What is stored in the frame trailer?
    A) IP address
    B) Error detection code
    C) File name
    D) User data

Answer: B

53. Which layer ensures reliable node-to-node communication?
    A) Network Layer
    B) Data Link Layer
    C) Physical Layer
    D) Application Layer

Answer: B

54. What happens when noise increases in a communication channel?
    A) Error rate decreases
    B) Error rate increases
    C) Bandwidth increases
    D) Data speed increases

Answer: B

55. What is the purpose of a repeater in networking?
    A) Convert protocols
    B) Extend network distance
    C) Encrypt data
    D) Manage routing

Answer: B

56. Why is flow control necessary in data communication?
    A) Prevent receiver overload
    B) Increase encryption

C) Reduce noise
D) Increase cable length

Answer: A

57. Which error detection technique is effective for detecting burst errors?
    A) Parity Check
    B) CRC
    C) ASCII
    D) Binary

Answer: B

58. Which function is mainly performed by the Data Link Layer?
    A) Routing
    B) Framing
    C) Encryption
    D) Compression

Answer: B

59. Which addressing method is used by the Data Link Layer?
    A) IP Address
    B) MAC Address
    C) Domain Name
    D) Port Number

Answer: B

60. Which of the following tasks is managed by the Data Link Layer?
    A) Web browsing
    B) Framing, addressing, and error detection
    C) File compression
    D) Database management

Answer: B

61. Which layer ensures reliable node-to-node communication?
    A) Physical Layer
    B) Data Link Layer
    C) Network Layer
    D) Transport Layer

Answer: B

62. Which unit of data is used at the Data Link Layer?
    A) Bit
    B) Packet
    C) Frame
    D) Segment

Answer: C

63. Which type of error occurs when unwanted signals affect data transmission?
    A) Frame error
    B) Noise error
    C) Protocol error
    D) Address error

Answer: B

64. What is the main function of a repeater?
    A) Routing data
    B) Regenerating signals
    C) Encrypting data
    D) Storing frames

Answer: B

65. Which topology provides multiple paths between devices?
    A) Bus
    B) Star
    C) Mesh
    D) Ring

Answer: C

66. What happens when two devices send data at the same time on a shared channel?
    A) Transmission success
    B) Collision occurs
    C) Data encrypted
    D) Data compressed

Answer: B

67. Which method is used to avoid collisions in Ethernet networks?
    A) ARQ
    B) CSMA/CD

C) Token passing
D) Sliding window

Answer: B

68. Which device filters frames based on MAC address?
    A) Hub
    B) Bridge
    C) Router
    D) Repeater

Answer: B

69. Which network type connects devices within a building?
    A) LAN
    B) WAN
    C) MAN
    D) PAN

Answer: A

70. Which layer is responsible for physical addressing?
    A) Network Layer
    B) Data Link Layer
    C) Transport Layer
    D) Application Layer

Answer: B

71. What is the main cause of transmission errors in networks?
    A) High bandwidth
    B) Noise
    C) Protocol
    D) Topology

Answer: B

72. Which device connects different networks together?
    A) Router
    B) Hub
    C) Switch
    D) Bridge

Answer: A

73. Which technique allows multiple signals to share a single communication channel?
    A) Routing
    B) Multiplexing
    C) Switching
    D) Encoding

Answer: B

74. Which standard defines wireless LAN technology?
    A) IEEE 802.3
    B) IEEE 802.5
    C) IEEE 802.11
    D) IEEE 802.1

Answer: C

75. Which topology uses a single backbone cable?
    A) Bus
    B) Star
    C) Mesh
    D) Ring

Answer: A

76. What does flow control prevent?
    A) Data duplication
    B) Receiver overload
    C) Signal delay
    D) Encryption errors

Answer: B

77. Which protocol requests retransmission of corrupted frames?
    A) ARQ
    B) FTP
    C) HTTP
    D) SMTP

Answer: A

78. What type of network covers an entire city?
    A) LAN
    B) MAN

C) PAN
D) SAN

Answer: B

79. Which device broadcasts incoming frames to all ports?
    A) Switch
    B) Router
    C) Hub
    D) Bridge

Answer: C

80. Which addressing method identifies a device on a network interface?
    A) IP Address
    B) MAC Address
    C) Port Number
    D) Domain Name

Answer: B

81. Which type of delay occurs due to signal travel time?
    A) Processing delay
    B) Propagation delay
    C) Transmission delay
    D) Queue delay

Answer: B

82. Which field in a frame is used to detect transmission errors?
    A) Header
    B) Data
    C) Trailer
    D) Address

Answer: C

83. What does CRC stand for?
    A) Cyclic Redundancy Check
    B) Central Routing Code
    C) Communication Rate Control
    D) Cyclic Routing Channel

Answer: A

84. Which topology is easiest to expand?
    A) Star
    B) Bus
    C) Ring
    D) Mesh

Answer: A

85. Which network device directs packets using logical addressing?
    A) Router
    B) Switch
    C) Bridge
    D) Hub

Answer: A

86. Which technique organizes data into frames before transmission?
    A) Framing
    B) Routing
    C) Switching
    D) Encoding

Answer: A

87. Which type of error detection uses polynomial division?
    A) Parity Check
    B) CRC
    C) Checksum
    D) ASCII

Answer: B

88. Which network topology has a central control device?
    A) Bus
    B) Star
    C) Mesh
    D) Ring

Answer: B

89. Which network device operates mainly at the data link layer?
    A) Switch
    B) Router

C) Gateway
D) Modem

Answer: A

90. Which topology becomes completely disconnected if the backbone cable fails?
    A) Star
    B) Bus
    C) Mesh
    D) Ring

Answer: B

91. Which process allows devices to share a communication channel?
    A) Access control
    B) Routing
    C) Encryption
    D) Encoding

Answer: A

92. Which network type connects personal devices within a few meters?
    A) LAN
    B) PAN
    C) WAN
    D) MAN

Answer: B

93. Which device translates between different network protocols?
    A) Router
    B) Gateway
    C) Switch
    D) Hub

Answer: B

94. Which component is responsible for sending data in communication?
    A) Receiver
    B) Sender
    C) Protocol
    D) Medium

Answer: B

95. Which network component receives the transmitted message?
    A) Sender
    B) Protocol
    C) Receiver
    D) Medium

Answer: C

96. Which factor determines how fast data can be transmitted?
    A) Noise
    B) Bandwidth
    C) Topology
    D) Protocol

Answer: B

97. Which unit represents data at the physical layer?
    A) Bit
    B) Frame
    C) Packet
    D) Segment

Answer: A

98. Which addressing type is permanent and assigned by manufacturer?
    A) IP Address
    B) MAC Address
    C) Domain Name
    D) Port Number

Answer: B

99. Which topology provides the highest fault tolerance?
    A) Bus
    B) Ring
    C) Mesh
    D) Star

Answer: C

100. Which layer is responsible for framing, physical addressing, and error detection?
    A) Network Layer
    B) Data Link Layer

C) Transport Layer
D) Application Layer

Answer: B

101. The Network Layer is the _____ layer of the OSI model.
A) First
B) Second
C) Third
D) Fourth

Answer: C

102. The primary function of the Network Layer is:
A) Routing packets
B) Framing
C) Encryption
D) Error detection

Answer: A

103. Data unit at the Network Layer is called:
A) Frame
B) Packet
C) Segment
D) Bit

104. Which device operates at the Network Layer?
A) Switch
B) Hub
C) Router
D) Repeater

Answer: C

105. The Network Layer is responsible for:
A) Logical addressing
B) Routing
C) Path determination
D) All of these

Answer: D

106. IP stands for:
    A) Internet Protocol
    B) Internal Protocol
    C) Internet Process
    D) Interface Protocol

Answer: A

107. IPv4 address length is:
    A) 16 bits
    B) 32 bits
    C) 64 bits
    D) 128 bits

Answer: B

108. IPv6 address length is:
    A) 32 bits
    B) 64 bits
    C) 128 bits
    D) 256 bits

Answer: C

109. An IP address identifies:
    A) Network cable
    B) Device on a network
    C) Router port
    D) Switch port

Answer: B

110. IPv4 address is divided into:
    A) Network and host parts
    B) Frame and packet parts
    C) Sender and receiver parts
    D) Header and trailer

Answer: A

111. Class A IP address range starts with:
    A) 1–126
    B) 128–191

C) 192–223
D) 224–239

Answer: A

112. Class B IP address range is:
A) 1–126
B) 128–191
C) 192–223
D) 224–239

Answer: B

113. Class C IP address range is:
A) 1–126
B) 128–191
C) 192–223
D) 224–239

Answer: C

114. Class D IP addresses are used for:
A) Multicasting
B) Private networks
C) Public networks
D) Broadcasting

Answer: A

115. Class E IP addresses are reserved for:
A) Research
B) Private networks
C) Multicast
D) Broadcast

Answer: A

116. Subnetting is used to:
A) Divide network into smaller networks
B) Increase bandwidth
C) Encrypt data
D) Compress data

Answer: A

117. Subnet mask is used to:
    A) Identify network and host parts
    B) Encrypt IP address
    C) Increase speed
    D) Reduce traffic

Answer: A

118. Default subnet mask of Class C is:
    A) 255.0.0.0
    B) 255.255.0.0
    C) 255.255.255.0
    D) 255.255.255.255

Answer: C

119. Default subnet mask of Class B is:
    A) 255.0.0.0
    B) 255.255.0.0
    C) 255.255.255.0
    D) 255.255.255.255

Answer: B

120. Default subnet mask of Class A is:
    A) 255.0.0.0
    B) 255.255.0.0
    C) 255.255.255.0
    D) 255.255.255.255

Answer: A

121. Routing means:
    A) Selecting path for data transmission
    B) Encrypting packets
    C) Compressing packets
    D) Framing data

Answer: A

122. Routing is performed by:
    A) Switch
    B) Router

C) Hub
D) Bridge

Answer: B

123. Routing table stores:
A) Path information
B) MAC address
C) Passwords
D) File data

Answer: A

124. Static routing means:
A) Manually configured routes
B) Automatic routing
C) Dynamic routing
D) Wireless routing

Answer: A

125. Dynamic routing means:
A) Automatic route updates
B) Manual routes
C) Static configuration
D) No routing

Answer: A

126. RIP stands for:
A) Routing Information Protocol
B) Router Internet Protocol
C) Routing Interface Protocol
D) Random Internet Protocol

Answer: A

127. OSPF stands for:
A) Open Shortest Path First
B) Online Secure Protocol Framework
C) Open System Packet Framework
D) Optical Switching Protocol Format

Answer: A

128. BGP stands for:
    A) Border Gateway Protocol
    B) Basic Gateway Protocol
    C) Binary Gateway Protocol
    D) Bridge Gateway Protocol

Answer: A

129. RIP uses:
    A) Distance vector routing
    B) Link state routing
    C) Static routing
    D) Wireless routing

Answer: A

130. OSPF uses:
    A) Link state routing
    B) Distance vector routing
    C) Static routing
    D) Random routing

Answer: A

131. Packet forwarding means:
    A) Sending packets to next hop
    B) Encrypting packets
    C) Compressing packets
    D) Storing packets

Answer: A

132. Next hop refers to:
    A) Next router in path
    B) Next computer
    C) Next server
    D) Next protocol

Answer: A

133. TTL in IP packet stands for:
    A) Time To Live
    B) Total Time Limit

C) Transfer Time Level
D) Transmission Time Line

Answer: A

134. TTL prevents:
A) Infinite looping of packets
B) Packet loss
C) Encryption
D) Compression

Answer: A

135. Packet header contains:
A) Address information
B) File data
C) Password
D) Compression data

Answer: A

136. ICMP stands for:
A) Internet Control Message Protocol
B) Internet Communication Management Protocol
C) Internal Control Message Process
D) Internet Channel Message Protocol

Answer: A

137. ICMP is used for:
A) Error reporting
B) File transfer
C) Email sending
D) Encryption

Answer: A

138. ARP stands for:
A) Address Resolution Protocol
B) Automatic Routing Protocol
C) Advanced Routing Process
D) Address Routing Protocol

Answer: A

139. ARP converts:
    A) IP address to MAC address
    B) MAC to IP
    C) IP to DNS
    D) DNS to IP

Answer: A

140. RARP stands for:
    A) Reverse Address Resolution Protocol
    B) Routing Address Resolution Protocol
    C) Remote Address Resolution Protocol
    D) Random Address Resolution Protocol

Answer: A

141. Network Layer ensures:
    A) End-to-end delivery of packets
    B) Node-to-node delivery
    C) File transfer
    D) Application communication

Answer: A

142. Logical addressing is provided by:
    A) Network Layer
    B) Data Link Layer
    C) Physical Layer
    D) Transport Layer

Answer: A

143. Router uses:
    A) Routing table
    B) MAC table
    C) ARP table only
    D) File table

Answer: A

144. Network congestion occurs when:
    A) Too many packets in network
    B) Router fails

C) Switch fails
D) Cable breaks

Answer: A

145. Network Layer protocols include:
A) IP
B) ICMP
C) ARP
D) All of these

Answer: D

146. Packet switching is used in:
A) Internet
B) Telephone networks
C) Radio networks
D) Television networks

Answer: A

147. IPv6 was introduced to:
A) Solve IPv4 address shortage
B) Increase encryption
C) Reduce bandwidth
D) Replace routers

Answer: A

148. Router selects path based on:
A) Routing algorithm
B) Encryption method
C) Data size
D) Cable type

Answer: A

149. Network Layer is responsible for:
A) Inter-network communication
B) Data encryption
C) File compression
D) Hardware control

Answer: A

150. The main goal of the Network Layer is:
    A) Reliable packet delivery across networks
    B) Framing data
    C) Error detection
    D) File transfer

Answer: A

151. The Transport Layer is the _____ layer in the OSI model.
    A) Third
    B) Fourth
    C) Fifth
    D) Sixth

Answer: B

152. The main function of the Transport Layer is:
    A) End-to-end communication
    B) Routing
    C) Framing
    D) Encryption

Answer: A

153. Data unit at the Transport Layer is called:
    A) Packet
    B) Frame
    C) Segment
    D) Bit

Answer: C

154. The Transport Layer provides:
    A) Process-to-process communication
    B) Node-to-node communication
    C) Device-to-device communication
    D) Network-to-network communication

Answer: A

155. Which protocols operate at the Transport Layer?
    A) TCP and UDP
    B) IP and ARP

C) HTTP and FTP
D) DNS and SMTP

Answer: A

156. TCP stands for:
A) Transmission Control Protocol
B) Transfer Communication Protocol
C) Transmission Communication Process
D) Transfer Control Process

Answer: A

157. TCP is a:
A) Connection-oriented protocol
B) Connectionless protocol
C) Routing protocol
D) Encryption protocol

Answer: A

158. TCP ensures:
A) Reliable transmission
B) Fast routing
C) Data compression
D) Encryption

Answer: A

159. TCP uses:
A) Acknowledgment system
B) Packet switching only
C) Broadcasting
D) Multicasting

Answer: A

160. TCP guarantees:
A) Ordered data delivery
B) Unordered delivery
C) Random delivery
D) Broadcast delivery

Answer: A

161. UDP stands for:
    A) User Datagram Protocol
    B) Universal Data Protocol
    C) User Data Process
    D) Universal Datagram Process

Answer: A

162. UDP is:
    A) Connectionless protocol
    B) Connection-oriented protocol
    C) Reliable protocol
    D) Routing protocol

Answer: A

163. UDP provides:
    A) Fast transmission
    B) Reliable delivery
    C) Error correction
    D) Flow control

Answer: A

164. Data unit in UDP is called:
    A) Segment
    B) Datagram
    C) Packet
    D) Frame

Answer: B

165. UDP is commonly used for:
    A) Streaming services
    B) Email
    C) File storage
    D) Data encryption

Answer: A

166. Port numbers are used to identify:
    A) Processes or applications
    B) Network cables

C) Routers
D) Switches

Answer: A

167. Port number size is:
A) 8 bits
B) 16 bits
C) 32 bits
D) 64 bits

Answer: B

168. Range of port numbers is:
A) 0–65535
B) 0–1024
C) 1–5000
D) 100–9999

Answer: A

169. Well-known ports range is:
A) 0–1023
B) 1024–49151
C) 49152–65535
D) 0–65535

Answer: A

170. Example of a well-known port:
A) Port 80
B) Port 3000
C) Port 6000
D) Port 50000

Answer: A

171. Flow control ensures:
A) Sender does not overwhelm receiver
B) Faster encryption
C) Data compression
D) Network routing

Answer: A

172. TCP flow control uses:
    A) Sliding window protocol
    B) Stop-and-Wait
    C) Token passing
    D) CSMA/CD

Answer: A

173. Window size determines:
    A) Amount of data sent before acknowledgment
    B) Network speed
    C) Cable length
    D) Router capacity

Answer: A

174. Sliding window protocol improves:
    A) Efficiency of transmission
    B) Encryption
    C) Network routing
    D) Compression

Answer: A

175. Flow control prevents:
    A) Buffer overflow
    B) Encryption failure
    C) Routing error
    D) Network failure

Answer: A

176. . Error control ensures:
    A) Reliable communication
    B) Fast routing
    C) Encryption
    D) Compression

Answer: A

177. TCP detects errors using:
    A) Checksum
    B) CRC

C) Parity bit
D) Encryption key

Answer: A

178. When an error occurs, TCP:
A) Retransmits the segment
B) Deletes data
C) Stops communication
D) Compresses data

Answer: A

179. Acknowledgment (ACK) indicates:
A) Successful receipt of data
B) Data loss
C) Routing failure
D) Encryption success

Answer: A

180. Negative acknowledgment means:
A) Data error occurred
B) Data received correctly
C) Network speed increased
D) Router failure

Answer: A

181. Congestion occurs when:
A) Network traffic exceeds capacity
B) Network is idle
C) Cable is disconnected
D) Router is turned off

Answer: A

182. TCP congestion control helps:
A) Reduce network traffic
B) Increase encryption
C) Compress data
D) Change IP address

Answer: A

183. Slow Start algorithm is used in:
    A) TCP congestion control
    B) UDP communication
    C) Routing protocols
    D) Encryption methods

Answer: A

184. Congestion window controls:
    A) Amount of data transmitted
    B) Network security
    C) Router speed
    D) Data compression

Answer: A

185. Congestion control improves:
    A) Network performance
    B) Data encryption
    C) Data storage
    D) File size

Answer: A

186. TCP connection establishment uses:
    A) Three-way handshake
    B) Two-way handshake
    C) One-way handshake
    D) Four-way handshake

Answer: A

187. Three-way handshake involves:
    A) SYN, SYN-ACK, ACK
    B) ACK, ACK, ACK
    C) SYN, ACK
    D) FIN, ACK

Answer: A

188. Connection termination uses:
    A) Four-way handshake
    B) Three-way handshake

C) Two-way handshake
D) One-way handshake

Answer: A

189. SYN packet is used to:
A) Start connection
B) End connection
C) Send data
D) Encrypt data

Answer: A

190. FIN packet is used to:
A) Terminate connection
B) Start communication
C) Encrypt data
D) Compress data

Answer: A

191. Transport Layer provides:
A) End-to-end delivery
B) Node-to-node delivery
C) Device-to-device delivery
D) Network-to-network delivery

Answer: A

192. Transport Layer sits between:
A) Network layer and Session layer
B) Physical layer and Data link layer
C) Application layer and Presentation layer
D) Data link layer and Physical layer

Answer: A

193. UDP does not provide:
A) Reliability
B) Speed
C) Fast communication
D) Low overhead

Answer: A

194. TCP header contains:
    A) Port numbers
    B) Sequence numbers
    C) Acknowledgment numbers
    D) All of these

Answer: D

195. Sequence numbers help in:
    A) Maintaining correct order of data
    B) Increasing bandwidth
    C) Encrypting data
    D) Compressing files

Answer: A

196. Transport Layer protocols work between:
    A) Applications
    B) Routers
    C) Switches
    D) Cables

Answer: A

197. TCP is preferred for:
    A) Reliable data transfer
    B) Fast streaming
    C) Broadcasting
    D) Multicasting

Answer: A

198. UDP is preferred for:
    A) Real-time applications
    B) File transfer
    C) Email communication
    D) Secure communication

Answer: A

199. Transport Layer provides:
    A) Multiplexing and demultiplexing
    B) Framing

C) Routing
D) Physical transmission

Answer: A

200. The main goal of the Transport Layer is:
A) Reliable end-to-end communication between processes
B) Routing packets
C) Physical transmission of data
D) Data encryption

Answer: A

201. The Application Layer is the _____ layer of the OSI model.
A) Fifth
B) Sixth
C) Seventh
D) Fourth

Answer: C

202. The Application Layer provides services directly to:
A) End users
B) Routers
C) Switches
D) Cables

Answer: A

203. Which OSI layer interacts with the user?
A) Network Layer
B) Application Layer
C) Data Link Layer
D) Physical Layer

Answer: B

204. The Application Layer is closest to the:
A) User
B) Hardware
C) Router
D) Switch

Answer: A

205. Which layer provides network services to applications?
    A) Transport Layer
    B) Application Layer
    C) Physical Layer
    D) Data Link Layer

Answer: B

206. HTTP stands for:
    A) Hyper Text Transfer Protocol
    B) High Text Transfer Protocol
    C) Hyper Terminal Transfer Protocol
    D) Hyperlink Text Transfer Protocol

Answer: A

207. TTP is used for:
    A) Web communication
    B) Email
    C) File transfer
    D) Routing

Answer: A

208. Default port number for HTTP is:
    A) 20
    B) 21
    C) 80
    D) 443

Answer: C

209. Secure version of HTTP is:
    A) HTTPS
    B) SHTTP
    C) HTTS
    D) HTCP

Answer: A

210. HTTPS uses port:
    A) 80
    B) 21

C) 443
D) 25

Answer: C

211. FTP stands for:
A) File Transfer Protocol
B) Fast Transfer Protocol
C) File Transmission Process
D) Fast Transmission Process

Answer: A

212. FTP is used to:
A) Transfer files
B) Send emails
C) Route packets
D) Encrypt data

Answer: A

213. FTP uses port number:
A) 20 and 21
B) 80
C) 25
D) 53

Answer: A

214. FTP works on:
A) Client-server model
B) Peer-to-peer
C) Broadcast network
D) Ring network

Answer: A

215. Anonymous FTP allows:
A) Public access without account
B) Secure login only
C) Router configuration
D) Email transfer

Answer: A

216. SMTP stands for:
    A) Simple Mail Transfer Protocol
    B) Secure Mail Transfer Protocol
    C) System Mail Transfer Protocol
    D) Standard Mail Transmission Process

Answer: A

217. SMTP is used for:
    A) Sending emails
    B) Receiving emails
    C) Routing emails
    D) Encrypting emails

Answer: A

218. Default SMTP port is:
    A) 25
    B) 21
    C) 80
    D) 53

Answer: A

219. POP3 stands for:
    A) Post Office Protocol version 3
    B) Packet Office Protocol 3
    C) Private Office Protocol 3
    D) Post Online Protocol 3

Answer: A

220. POP3 is used to:
    A) Retrieve email from server
    B) Send email
    C) Route packets
    D) Encrypt data

Answer: A

221. IMAP stands for:
    A) Internet Message Access Protocol
    B) Internet Mail Access Process

C) Internal Mail Access Protocol
D) Internet Message Authentication Protocol

Answer: A

222. IMAP allows users to:
A) Access email on server
B) Send emails only
C) Transfer files
D) Encrypt emails

Answer: A

223. Default IMAP port number is:
A) 143
B) 25
C) 21
D) 80

Answer: A

224. IMAP is better than POP3 because:
A) Emails remain on server
B) Faster routing
C) Better encryption
D) Smaller packets

Answer: A

225. Secure IMAP uses port:
A) 993
B) 25
C) 80
D) 20

Answer: A

226. DNS stands for:
A) Domain Name System
B) Data Name System
C) Domain Network System
D) Data Network Service

Answer: A

227. DNS converts:
    A) Domain name to IP address
    B) IP to MAC
    C) MAC to IP
    D) Data to packet

Answer: A

228. DNS uses port number:
    A) 53
    B) 21
    C) 80
    D) 25

Answer: A

229. Example of a domain name:
    A) www.google.com
    B) 192.168.1.1
    C) 255.255.255.0
    D) 10.0.0.1

Answer: A

230. DNS servers store:
    A) Domain-IP mappings
    B) Passwords
    C) Emails
    D) Files

Answer: A

231. DHCP stands for:
    A) Dynamic Host Configuration Protocol
    B) Data Host Configuration Process
    C) Dynamic Hardware Configuration Protocol
    D) Domain Host Communication Protocol

Answer: A

232. DHCP automatically assigns:
    A) IP addresses
    B) MAC addresses

C) Email addresses
D) Port numbers

Answer: A

233. DHCP reduces:
A) Manual network configuration
B) Network speed
C) Data transfer
D) Encryption

Answer: A

234. DHCP works on:
A) Client-server model
B) Peer-to-peer
C) Broadcast only
D) Token passing

Answer: A

235. DHCP uses ports:
A) 67 and 68
B) 20 and 21
C) 80 and 443
D) 25 and 110

Answer: A

236. Telnet is used for:
A) Remote login
B) Email transfer
C) File transfer
D) DNS resolution

Answer: A

237. Default port for Telnet is:
A) 23
B) 80
C) 21
D) 25

Answer: A

238. SSH stands for:
    A) Secure Shell
    B) Secure System Host
    C) System Secure Host
    D) Shell Security Host

Answer: A

239. SSH provides:
    A) Secure remote access
    B) File compression
    C) Email transfer
    D) Packet routing

Answer: A

240. Default SSH port is:
    A) 22
    B) 21
    C) 80
    D) 25

Answer: A

241. Web browsers use:
    A) HTTP
    B) FTP
    C) SMTP
    D) ARP

Answer: A

242. Email services use:
    A) SMTP, POP3, IMAP
    B) FTP and DNS
    C) HTTP and ARP
    D) RIP and OSPF

Answer: A

243. DNS improves:
    A) Internet usability
    B) Network speed

C) Cable quality
D) Encryption

Answer: A

244. Application Layer protocols provide:
A) Network services to users
B) Routing services
C) Data encryption only
D) Physical data transmission

Answer: A

245. Application Layer protocols include:
A) HTTP
B) FTP
C) DNS
D) All of these

Answer: D

246. Application Layer is responsible for:
A) User interface to network services
B) Packet routing
C) Data framing
D) Signal transmission

Answer: A

247. Application Layer protocols are used by:
A) Software applications
B) Routers
C) Switches
D) Network cables

Answer: A

248. Example of Application Layer service:
A) Web browsing
B) Packet forwarding
C) Signal transmission
D) Framing

Answer: A

249. Email communication requires:
   A) SMTP and POP3/IMAP
   B) HTTP and FTP
   C) RIP and OSPF
   D) ARP and RARP

Answer: A

250. The main goal of the Application Layer is:
   A) Provide network services directly to applications and users
   B) Route packets
   C) Detect errors
   D) Control hardware

Answer: A

251. Wireless Networks) Which IEEE standard defines Wi-Fi networks?
   A) IEEE 802.3
   B) IEEE 802.11
   C) IEEE 802.15
   D) IEEE 802.16

Answer: B

252. Which frequency bands are commonly used by Wi-Fi?
   A) 1 GHz and 2 GHz
   B) 2.4 GHz and 5 GHz
   C) 900 MHz only
   D) 10 GHz

Answer: B

253. Bluetooth technology is based on which IEEE standard?
   A) 802.15
   B) 802.11
   C) 802.16
   D) 802.3

Answer: A

254. Which wireless network type covers the smallest area?
   A) WLAN
   B) WPAN

C) WMAN
D) WWAN

Answer: B

255. Which protocol is used by Wi-Fi for medium access?
A) CSMA/CD
B) CSMA/CA
C) TDMA
D) FDMA

Answer: B

256. Which device connects wireless users to a wired LAN?
A) Router
B) Access Point
C) Switch
D) Repeater

Answer: B

257. WiMAX is defined by which IEEE standard?
A) 802.16
B) 802.11
C) 802.3
D) 802.5

Answer: A

258. ZigBee is mainly designed for:
A) High speed multimedia
B) IoT and sensor networks
C) Satellite communication
D) Optical networking

Answer: B

259. The coverage region of a base station is called:
A) Cluster
B) Node
C) Cell
D) Sector

Answer: C

260. Movement of a mobile device between cells is called:
    A) Roaming
    B) Routing
    C) Switching
    D) Fragmentation

Answer: A

261. Transfer of connection between cells is called:
    A) Handoff
    B) Broadcast
    C) Encryption
    D) Multiplexing

Answer: A

262. LTE is associated with which generation?
    A) 2G
    B) 3G
    C) 4G
    D) 1G

Answer: C

263. Spread spectrum improves:
    A) Security and interference resistance
    B) Packet delay
    C) Storage
    D) Power usage

Answer: A

264. Typical Bluetooth range is approximately:
    A) 10 meters
    B) 1 km
    C) 100 km
    D) 500 meters

Answer: A

265. Wi-Fi 6 corresponds to:
    A) 802.11g
    B) 802.11n

C) 802.11ac
D) 802.11ax

Answer: D

266. OFDM stands for:
A) Orthogonal Frequency Division Multiplexing
B) Optical Frequency Division Method
C) Open Frequency Division Method
D) Orthogonal Frame Division Method

Answer: A

267. An ad-hoc wireless network:
A) Uses a central router
B) Requires satellite
C) Has no fixed infrastructure
D) Works only with fiber

Answer: C

268. RSSI measures:
A) Data speed
B) Signal strength
C) Latency
D) Packet size

Answer: B

269. Which problem occurs when nodes cannot detect each other's transmission?
A) Hidden node problem
B) Broadcast storm
C) Routing loop
D) Fragmentation

Answer: A

270. MIMO technology improves:
A) Bandwidth and throughput
B) Battery usage
C) Cable length
D) Storage capacity

Answer: A

271. NFC communication range is typically:
    A) Few centimeters
    B) 10 km
    C) 100 m
    D) 1 km

Answer: A

272. Beacon frames are used for:
    A) Network synchronization
    B) Encryption
    C) Routing
    D) Compression

Answer: A

273. DSSS stands for:
    A) Direct Sequence Spread Spectrum
    B) Dynamic Signal Spread System
    C) Distributed Signal Switching System
    D) Data Spread Spectrum System

Answer: A

274. Which network supports self-healing paths?
    A) Mesh network
    B) Bus network
    C) Star network
    D) Ring network

Answer: A

275. Multipath fading occurs due to:
    A) Multiple signal reflections
    B) Packet encryption
    C) Routing loops
    D) Hardware failure

Answer: A

276. Wi-Fi hotspot provides:
    A) Wireless internet access
    B) Satellite connectivity

C) Optical networking
D) Cable routing

Answer: A

277. IEEE 802.11g maximum theoretical speed:
A) 11 Mbps
B) 54 Mbps
C) 100 Mbps
D) 600 Mbps

Answer: B

278. Channel bonding increases:
A) Bandwidth
B) Packet loss
C) Interference
D) Power consumption

Answer: A

279. Which technique allows multiple users to share a channel?
A) Multiplexing
B) Routing
C) Encryption
D) Compression

Answer: A

280. ZigBee is known for:
A) Low power consumption
B) High speed video
C) Large bandwidth
D) Satellite communication

Answer: A

281. In wireless communication, interference mainly affects:
A) Signal quality
B) File storage
C) Disk speed
D) Printer performance

Answer: A

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D) Printer performance

Answer: A

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B) File storage
C) Disk speed
D) Printer performance

Answer: A

291. Cloud computing provides services over:
A) Internet
B) USB
C) LAN only
D) Satellite only

Answer: A

292. IaaS stands for:
A) Infrastructure as a Service
B) Internet as a Service
C) Integration as a Service
D) Instance as a Service

Answer: A

293. PaaS mainly provides:
    A) Development platform
    B) Network cables
    C) Physical routers
    D) Operating systems only

Answer: A

294. SaaS delivers:
    A) Software applications online
    B) Hardware devices
    C) Network switches
    D) Optical fibers

Answer: A

295. Which company provides major cloud services?
    A) Amazon
    B) Intel
    C) Cisco
    D) Oracle hardware

Answer: A

296. Virtualization allows:
    A) Multiple VMs on one machine
    B) Only one OS per machine
    C) Hardware removal
    D) No operating system

Answer: A

297. Hypervisor is used to:
    A) Manage virtual machines
    B) Encrypt files
    C) Compress data
    D) Route packets

Answer: A

298. Google Docs is example of:
    A) SaaS
    B) IaaS

C) PaaS
D) DaaS

Answer: A

299. Cloud scalability refers to:
A) Ability to increase resources
B) Removing security
C) Increasing latency
D) Decreasing reliability

Answer: A

300. Elasticity means:
A) Automatic scaling of resources
B) Manual hardware upgrade
C) Slower network
D) Reduced storage

Answer: A

301. Which of the following is a primary characteristic of cloud computing?
A) Limited scalability
B) On-demand self-service
C) Dedicated hardware ownership
D) Manual provisioning

Answer: B

302. Which cloud service model provides virtual machines, storage, and networking resources?
A) SaaS
B) PaaS
C) IaaS
D) DaaS

Answer: C

303. Which deployment model is operated solely for a single organization?
A) Public Cloud
B) Private Cloud
C) Hybrid Cloud
D) Community Cloud

Answer: B

304. Which organization developed the widely accepted definition of cloud computing?
    A) IEEE
    B) NIST
    C) ISO
    D) ITU

Answer: B

305. Which cloud service model provides a ready-to-use application over the internet?
    A) SaaS
    B) IaaS
    C) PaaS
    D) NaaS

Answer: A

306. Which technology is primarily used to enable virtualization in cloud computing?
    A) Hypervisor
    B) Compiler
    C) Assembler
    D) Interpreter

Answer: A

307. Which of the following is an example of a public cloud provider?
    A) Microsoft Azure
    B) Private Datacenter
    C) Local Server
    D) Personal Computer

Answer: A

308. Which feature allows cloud resources to automatically scale up or down?
    A) Elasticity
    B) Encryption
    C) Authentication
    D) Replication

Answer: A

309. Which cloud model combines public and private clouds?
    A) Hybrid Cloud

B) Private Cloud
C) Community Cloud
D) Edge Cloud

Answer: A

310. Which storage service stores unstructured data in the cloud?
A) Block Storage
B) Object Storage
C) File Storage
D) Local Storage

Answer: B

311. The CIA triad includes:
A) Confidentiality Integrity Availability
B) Control Internet Access
C) Centralized Information Access
D) Communication Integrity Algorithm

Answer: A

312. Encryption converts:
A) Plaintext to ciphertext
B) Ciphertext to plaintext
C) Packets to frames
D) Frames to bits

Answer: A

313. AES is:
A) Symmetric encryption algorithm
B) Hash function
C) Routing protocol
D) Compression algorithm

Answer: A

314. RSA is:
A) Asymmetric encryption
B) Symmetric encryption
C) Hashing method
D) Firewall rule

Answer: A

315. Firewall is used to:
    A) Control network traffic
    B) Compress data
    C) Store files
    D) Increase bandwidth

Answer: A

316. IDS stands for:
    A) Intrusion Detection System
    B) Internet Data Server
    C) Internal Device Service
    D) Integrated Data System

Answer: A

317. VPN provides:
    A) Secure remote access
    B) Faster CPU
    C) Disk storage
    D) Wireless coverage

Answer: A

318. Phishing attack attempts to:
    A) Steal sensitive information
    B) Increase bandwidth
    C) Improve encryption
    D) Fix vulnerabilities

Answer: A

319. Malware includes:
    A) Virus worm trojan
    B) Router switch hub
    C) HTML CSS JS
    D) LAN MAN WAN

Answer: A

320. DDoS attack aims to:
    A) Overload service

B) Improve routing
C) Encrypt files
D) Authenticate users

Answer: A

321. HTTPS provides security using:
A) TLS/SSL encryption
B) FTP protocol
C) SMTP protocol
D) Telnet

Answer: A

322. NGN stands for:
A) Next Generation Network
B) Network Global Node
C) New Grid Network
D) Node Generation Network

Answer: A

323. NGN is primarily based on:
A) IP technology
B) Analog switching
C) Circuit switching only
D) Satellite only

Answer: A

324. VoIP means:
A) Voice over Internet Protocol
B) Video over Internet Port
C) Voice Optical Interface
D) Virtual Output IP

Answer: A

325. SIP protocol is used for:
A) Session initiation
B) Data compression
C) Packet filtering
D) Encryption

Answer: A

326. IMS stands for:
   A) IP Multimedia Subsystem
   B) Internet Messaging Server
   C) Integrated Mobile System
   D) Intelligent Management Server

Answer: A

327. 5G networks provide:
   A) High speed and low latency
   B) Only voice calls
   C) No internet
   D) Satellite only

Answer: A

328. Network slicing is mainly used in:
   A) 5G networks
   B) 2G networks
   C) Ethernet LAN
   D) DSL networks

Answer: A

329. Which security goal ensures that information is not disclosed to unauthorized users?
   A) Integrity
   B) Confidentiality
   C) Availability
   D) Authentication

Answer: B

330. Which security goal ensures that data is accurate and not modified?
   A) Integrity
   B) Confidentiality
   C) Availability
   D) Authorization

Answer: A

331. Which type of attack attempts to make a service unavailable to users?
   A) Phishing
   B) DoS Attack

C) Man-in-the-middle
D) Spoofing

Answer: B

332. Which protocol is commonly used to securely browse websites?
A) HTTP
B) FTP
C) HTTPS
D) SMTP

Answer: C

333. Which technique hides the content of a message to protect it from unauthorized access?
A) Compression
B) Encryption
C) Fragmentation
D) Encoding

Answer: B

334. Which type of malware replicates itself without attaching to other programs?
A) Worm
B) Trojan
C) Virus
D) Spyware

Answer: A

335. Which authentication method uses two different verification factors?
A) Single-factor authentication
B) Multi-factor authentication
C) Password authentication
D) Token authentication

Answer: B

336. Which network device monitors and filters incoming and outgoing network traffic based on security rules?
A) Router
B) Switch
C) Firewall
D) Repeater

Answer: C

337. Which attack involves intercepting communication between two parties?
   A) Man-in-the-middle attack
   B) DoS attack
   C) Brute force attack
   D) Dictionary attack

Answer: A

338. Which protocol is used for secure remote login to servers?
   A) Telnet
   B) SSH
   C) FTP
   D) SNMP

Answer: B

339. Which attack involves sending fraudulent emails to trick users into revealing sensitive information?
   A) Phishing
   B) Sniffing
   C) Spoofing
   D) Injection

Answer: A

340. Which method ensures that a user cannot deny performing a specific action?
   A) Authentication
   B) Authorization
   C) Non-repudiation
   D) Confidentiality

Answer: C

341. Which network security technique monitors traffic to detect suspicious activities?
   A) IDS
   B) DHCP
   C) DNS
   D) NAT

Answer: A

342. Which encryption uses the same key for encryption and decryption?
   A) Asymmetric encryption
   B) Symmetric encryption

C) Hashing
D) Digital signature

Answer: B

343. Which encryption uses two different keys (public and private)?
A) Symmetric encryption
B) Asymmetric encryption
C) Hashing
D) Encoding

Answer: B

344. Which protocol secures email transmission?
A) SMTP
B) POP3
C) S/MIME
D) FTP

Answer: C

345. Which type of firewall filters traffic based on IP addresses and ports?
A) Packet filtering firewall
B) Proxy firewall
C) Application firewall
D) Stateful firewall

Answer: A

346. Which attack involves guessing passwords by systematically trying all possibilities?
A) Phishing
B) Brute force attack
C) DoS attack
D) Spoofing

Answer: B

347. Which security technique converts data into a fixed-length string for integrity verification?
A) Encryption
B) Hashing

C) Encoding
D) Compression

Answer: B

348. Which protocol provides network-layer security for IP communications?
A) SSL
B) TLS
C) IPSec
D) SSH

Answer: C